Consumed Methadone Dose Patterns Around Randomly Scheduled Urine Testing: Pre-Test Escalation and Higher-Risk Trajectories in Long-Retained Maintenance Patients

Authors: [AUTHOR LIST — name, degree, affiliation; designate corresponding author + email]

Affiliation: [INSTITUTION NAME AND ADDRESS]

Corresponding author: [NAME, EMAIL]

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ABSTRACT

Aims. To estimate whether short-term patterns of change in the consumed methadone dose around a urine test are associated with a concurrent morphine-positive result, and whether such patterns form a reproducible partition of dose-change shape.

Design. Retrospective analysis of routinely collected daily dosing records and urine test results, 2018–2022, reported following the STROBE statement.

Setting. A single methadone maintenance programme in which methadone was dispensed and consumed on site under staff observation, take-home dosing was not permitted, and urine collection occasions were randomly scheduled.

Participants. 552 patients contributing 9,900 analysable morphine urine tests, of which 3,401 were positive, drawn from a source cohort that the programme had restricted to patients retained for at least two years.

Measurements. Each test's daily consumed-dose series was expressed as a ratio to its own value seven days earlier; non-attended days had been interpolated upstream. Dose-change shape was summarised by the polar angles of the pre-test and post-test slopes and clustered on the circle using a mixture of von Mises distributions with K = 6. Associations with the urine result were estimated by logistic regression with patient-clustered standard errors and false-discovery-rate control.

Results. Nine patterns were identified, six of dose change and three of stable dosing, the latter exploratory, spanning positive/negative occurrence ratios from 1.64 to 0.50. Cluster membership persisted under outcome-blind re-clustering, after adjustment for attendance, in a within-patient comparison and on windows containing no interpolated day, and it was not reducible to any single continuous slope. On imputation-free windows pre-test escalation strengthened, with a within-patient odds ratio rising from 1.15 to 1.26 per standard deviation (95% CI 1.15–1.38, $p = 3\times10^{-7}$), whereas the post-test escalation magnitude became inconclusive (1.36 to 1.06, 95% CI 0.97–1.15, $p = 0.23$) and is not claimed. Attendance during the preceding week was the strongest single correlate, with an odds ratio of 0.84 per 0.1 increment ($p = 7\times10^{-22}$). The dose-change association did not replicate across a temporal split at April 2020.

Summary. Among long-retained methadone maintenance patients, the shape of consumed-dose change around a randomly scheduled urine test appears to be associated with the concurrent morphine result. The antecedent component is robust to imputation of non-attended days, while the post-test magnitude is not. The signal seems to add little or nothing predictive over attendance and prior testing history, and the findings describe long-retained maintenance patients only.

Keywords: methadone maintenance therapy; opioid use disorder; illicit opioid use; urine drug testing; dose trajectory; circular clustering; observational study.

INTRODUCTION

Methadone maintenance therapy is a cornerstone of treatment for opioid use disorder. Pooled trial evidence and a systematic review both find that it retains patients and reduces illicit opioid use [1,2], and the effect appears dose-dependent, in that daily doses of 60–100 mg improve retention and urine-verified abstinence relative to lower doses, and flexible dosing outperforms fixed dosing [3,4]. If the level of the dose matters in this way, the way a patient's dose moves from day to day may carry clinical information as well. Continued illicit-opioid use during maintenance, detected by routine urine drug testing, remains the principal marker of relapse and predicts attrition, a positive test roughly tripling the risk of leaving treatment [8].

Machine-learning models of opioid-treatment outcomes have been built from claims and electronic health-record features to predict retention or discontinuation [10,11], although external validation is rare and reporting often incomplete [9]. These models draw on aggregate prescribing, claims or note-level features and not on the shape of the daily dose series, and where the daily methadone dose has been modelled alongside urine opiate results it has been treated as an outcome to be optimised for prescribing [15] and not as a behavioural signal. The trajectory of the daily dose around a test may nevertheless be informative in itself, given that withdrawal, dose-seeking, missed doses and clinician-led adjustments would each leave a characteristic short-term signature.

Unsupervised methods have been applied to urine-use outcome sequences in addiction trials [12], to prescription-opioid dose trajectories in chronic-pain cohorts [13] and to daily methadone dispensing-state sequences in opioid treatment programmes [16], and dose-change pattern features have been related to outcome for tapers at the end of treatment [17]. What has not been examined is the short-term shape of methadone dose change around a urine test, taken as a marker of concurrent illicit-opioid use. That the peri-test window carries behavioural information is not itself a new idea, given that contingency schedules tying the dose to the urine result change urinalysis outcomes [18], but to our knowledge it has not been characterised observationally as a trajectory shape. We hypothesised that patients with positive urine tests show distinct patterns of dose change around the test, capturable in a reproducible partition of shape. Building on an earlier internal analysis of this cohort, we replaced visually drawn boundaries with model-based clustering that respects the circular geometry of the features [14], accounted for repeated testing within patients, and evaluated honest out-of-sample predictive value. The urine result is treated at face value; misclassification, well documented in maintenance treatment [5,6,7], is addressed under Limitations.

METHODS

Setting and data source

De-identified records were obtained from the methadone maintenance programme at [INSTITUTION NAME]. The daily dosing extract runs from January 2018 to 31 December 2022. The urine extract runs a further five months, but a test is analysable only if the dosing window around it contains an observed dose, so the study period is January 2018 to December 2022. Methadone was dispensed and consumed on site under staff observation on the day of attendance. Any portion the patient did not drink was returned and recorded, and take-home dosing was not permitted at any point, so the consumed dose is observed ingestion and not a proxy for it. The daily record holds the prescribed dose in milligrams and the consumed dose as a volume in cubic centimetres. The recorded volume falls below one tenth of the prescribed milligram dose on

71.5% of dosing days, but we do not interpret that shortfall behaviourally, because the extract does not state the syrup concentration and the milligram-to-volume ratio varies across records (interquartile range 10–15.7). The analysis therefore uses the consumed series normalised to its own value seven days earlier, which is invariant to a per-patient concentration.

Urine collection occasions were randomly scheduled and were not ordered in response to clinical suspicion, so pre-test dosing behaviour cannot have been elicited by the decision to test. Each test contributed a patient identifier, a test date and a qualitative morphine result. A subset of patients underwent supervised collection, which assures specimen validity but does not change the detection window of the assay, so an observed negative may still miss intermittent use [5,6]. These observed-collection negatives are used in a sensitivity analysis (Supplementary S1) as a higher-specificity comparator and not as a gold standard. The study concerns morphine only, and amphetamine results are reported separately.

Ethics

The study was approved by [ETHICS COMMITTEE NAME] (approval number [IRB NUMBER]). All data were de-identified before analysis, and the requirement for individual informed consent was waived for this retrospective analysis of routinely collected records.

Inclusion and feature construction

The analysable sample was defined by two rules. The cohort supplied by the programme comprised the 599 patients retained in maintenance treatment for at least two years, a restriction inherited from the extract and not chosen here. And a fixed 57-day window spanning day −28 to day +28 was constructed around each urine test, analysable if it contained at least one attended dosing day.

The window is a fixed calendar grid, so days on which the patient did not attend are missing by construction, and each such day was filled by linear interpolation between the nearest observed doses, with edge runs carried across, before any feature was computed. The imputation is not negligible, affecting 15.1% of ±7-day and 15.8% of ±28-day observations for positive tests and 7.9% and 8.7% respectively for negative tests, so it is treated as a primary analytic concern and not as a technicality (Table S5).

Each daily series was expressed as a ratio to its own value seven days before the test ($\tilde{d}_t = d_t / d_{-7}$, $t = -28 \ldots +28$), for the consumed and prescribed series alike. A three-day moving window advancing one day at a time across the seven days either side recorded the minimum and maximum window slopes on each side. The two escalation scalars used throughout are these maxima, taken over days −7 to 0 and days 0 to +7 respectively, both reported per within-sample standard deviation. Tests were then split into two branches.

- Dose-change branch, comprising tests whose dose varied within ±7 days. The (post-test, pre-test) minimum- and maximum-slope pairs were converted to polar coordinates, retaining the angles θ_{min} and θ_{max}, which makes the shape of a dose change comparable independent of its absolute magnitude.
- Stable-dose branch, comprising tests whose dose was unchanged within ±7 days. The ±28-day normalised consumed- and prescribed-dose series (114 variables) were reduced by principal-component analysis fitted on the positive tests, with negative tests projected onto the same axes, and the first two components (58.1% of variance) were converted to a single angle θ.

Clustering

The discriminating features are angles, so clusters were defined on the circle and not in Euclidean space, which does not respect wrap-around at 0° and 360° (Figure 1).

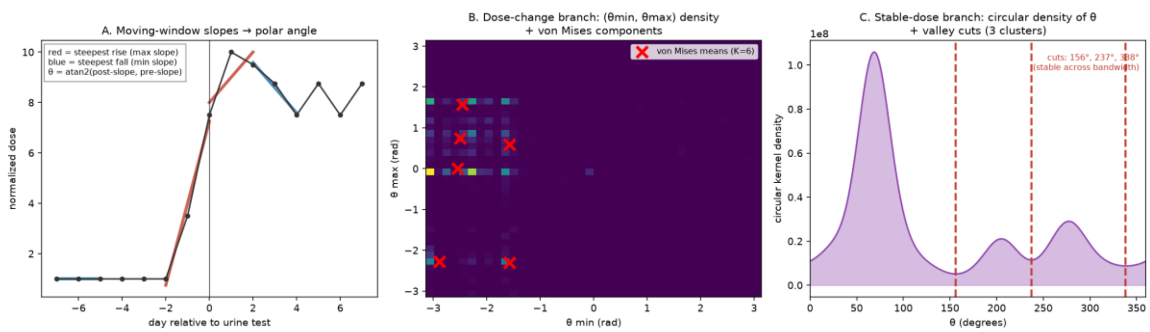

Figure 1. Method overview. (A) An example dose trajectory, in which three-day moving-window slopes before and after the test (steepest rise in red, steepest fall in blue) are combined into polar angles θ = atan2(post-slope, pre-slope). (B) Dose-change branch, showing the two-dimensional density of (θmin, θmax) with the six von Mises component means marked by a red ×. (C) Stable-dose branch, showing the circular kernel density of the principal-component angle θ with the three density-valley cut points at 156°, 237° and 338°.

In the dose-change branch, (θ_min, θ_max) form a toroidal continuum with no natural number of clusters, and neither the Bayesian nor the Integrated Completed Likelihood criterion reached a minimum as components were added. The number of components is therefore an investigator choice and not a statistical selection. We fixed a parsimonious, clinically interpretable $K = 6$ and fitted a mixture of bivariate (product) von Mises distributions by expectation-maximisation [14]. Given that K is a choice, the cluster-outcome pattern is reported across $K = 4$ to 8 (Table S4). In the stable-dose branch, θ showed three modes, and boundaries were placed at the minima of its circular kernel-density estimate (156°, 237° and 338.5°, stable across $\kappa = 8$ to 16). An alternative one-dimensional von Mises mixture with $K = 3$, fitted to the same tests, yields a different partition under which no stable-dose cluster is higher-risk (Table S6). That partition cannot be said to have been chosen blind to the difference, so the stable-dose branch is exploratory throughout.

The taxonomy was derived on the positive tests alone, on the reasoning that routine urine testing under-detects intermittent use [5,6] and the negative set therefore contains false negatives which would contaminate cluster definition; negatives were then assigned to the fitted clusters. Because that could make the analysis circular, the whole clustering was repeated outcome-blind on the pooled sample, and reproducibility was assessed by 500 bootstrap refits of each branch's own procedure (Table S3).

Statistical analysis

For each cluster the positive/negative occurrence ratio was computed as P/N = (cluster positives / 3,401) ÷ (cluster negatives / 6,499), a ratio of occurrence rates. Association between cluster membership and a positive result was tested by logistic regression with standard errors clustered by patient, as patients contributed multiple tests, and p-values were adjusted for the false discovery rate across clusters. Clusters with P/N above 1.2 and $q < 0.05$ were labelled higher-risk and those with P/N below 0.83 and $q < 0.05$ lower-risk; these thresholds are descriptive conventions and not findings. Within-patient estimates used conditional logistic regression with the patient as stratum.

Attendance in the week before the test is a proportion on 0 to 1. Every attendance odds ratio is reported per 0.1 increment and every dose-slope odds ratio per standard deviation, and pooled and within-patient estimates are always compared on the same scale. The study is associational and not predictive, though for completeness we report how much the dose trajectory adds over a baseline of attendance and prior testing history. There is no registered protocol and no pre-registered analysis plan, so all findings should be read as exploratory in that sense. Analyses used Python 3.12 (scikit-learn, statsmodels), implemented with assistance from a large language model and verified by re-running every script against the source records (see Declarations); the upstream 57-day windows and the interpolation were built by the authors in MATLAB.

RESULTS

Cohort and participant flow

Over 2018–2022 the programme recorded 20,108 morphine urine test records (6,830 positive and 13,278 negative) from 3,213 patients, of which 677 were repeat records for the same patient-day that the pipeline retained, giving 19,431 distinct patient-days. The funnel to the analysable sample was as follows.

Step	Tests	Patients	Reason for exclusion at this step	Excluded
1. Morphine urine test records, 2018–2022	20,108	3,213	—	—
2. Patient present in the daily methadone extract	15,592	1,509	No dosing record for that patient	4,516
3. Patient on the programme's ≥2-year retention list	10,378 (3,579 pos / 6,799 neg)	553	Retained in MMT <2 years (§2.3)	5,214
4. ≥1 attended dosing day inside the ±28-day window	**9,900** (**3,401 pos** / **6,499 neg**)	**552**	No observed dose anywhere in the window, so nothing to interpolate from	478 (178 pos / 300 neg)

The analysable sample comprises 3,401 morphine-positive tests from 472 patients and 6,499 morphine-negative tests from 531 patients, 552 unique patients in total, 451 of them contributing both outcomes. The funnel is reproduced end to end by `35_strobe_flow.py`, which re-derives every count from the source spreadsheets and fails if any drifts.

Two steps need explanation and not only a count. Step 2 loses 4,516 tests because the daily-dose extract covers only 1,672 patients against the 3,213 with a morphine result, for reasons not recoverable from the files supplied; the 1,704 unlinked patients contributed a median of 1 test each and had a markedly lower positive rate (0.184 versus 0.385, $\chi^2 = 630$, $p < 10^{-100}$), so their exclusion raises baseline positivity and the reported rate is not a programme-wide rate. Step 3 reaches 553 and not 599 patients because 13 of the supplied list lack a morphine urine record and 33 lack a dosing record.

Characteristics of the analysable cohort are given in Table 1. Age and sex are absent from every source file supplied and therefore cannot be reported, and recorded body weight was likewise unusable, the field being present but empty, and all dispensing occurred within a single programme.

Table 1. Baseline characteristics of the analysable cohort (552 patients, 9,900 urine tests), as medians with interquartile ranges computed per patient from source records (`41_baseline_table.py`).

Characteristic	Median (IQR)
Age, years	**not available in the data extract**
Sex	**not available in the data extract**
Duration of methadone record, months	58.5 (38.6–60.0)
Attended dosing days per patient	1,418 (989–1,731)
Prescribed methadone dose, mg/day	80.0 (60.0–100.0)
Consumed (dispensed) methadone dose, cc/day	6.0 (4.0–8.0)
Analysable morphine urine tests per patient	12 (7–29)
Morphine-positive tests per patient, %	32.5 (10.0–60.0)

The cohort is thus one of long-established maintenance patients, with a median of almost five years of dosing records and nearly 1,500 attended dosing days per patient, at doses (median 80 mg/day) within the range associated with adequate maintenance [3,4]. The wide interquartile range of per-patient positivity, from 10% to 60%, suggests substantial heterogeneity in illicit-opioid use despite this stability.

The largest exclusion is step 3, whose two-year retention restriction was imposed by the programme and not chosen here, and the excluded patients differ systematically. Tests from patients retained for less than two years had a higher positive rate (0.464 versus 0.345), much lower pre-test attendance (mean 0.489 versus 0.836) and far fewer tests per patient (median 3 versus 13). The findings therefore describe stably retained maintenance patients. Because the excluded occasions are concordantly low in attendance and high in positivity, excluding them restricts the range of the attendance variable and would be expected to attenuate rather than inflate the reported attendance association.

The dose-change partition and its ordering

Clustering produced nine clusters, six in the dose-change branch and three in the stable-dose branch. Reproducibility was quantified by 500 bootstrap refits of each branch's own procedure (Table S3, which also records a correction to an earlier version of this analysis).

Cluster boundaries move under resampling, with mean best-match Jaccard ranging from 0.32 to 0.77 and a branch adjusted Rand index of 0.62, as would be expected of a partition imposed on a continuum. Cluster risk seems better determined than a naive bootstrap suggests. Mixture components carry no fixed identity across refits, so per-cluster intervals must be taken after matching each refit's components to the original clusters, which narrows them by 29% (Table S3b). Two clusters then exclude unity (7d-2, 1.41–1.80; 7d-4, 1.05–2.31) and 7d-3 is reliably directional (0.42–1.08), while 7d-1, 7d-5 and 7d-6 remain imprecise. These intervals must be read against the behaviour of the same clusters elsewhere, because the two with the narrowest intervals fare worst in the other analyses: 7d-2 has no distinct outcome-blind counterpart, and 7d-4 is the one cluster whose within-patient estimate as a lone indicator is inconclusive (OR 1.12, $p = 0.15$).

The escalation ordering appears more stable than any individual boundary. A rank correlation across $K = 4$ to 8 is under-powered at every resolution (Table S4), so we recomputed it on each of the 500 refits, ranking components by mean pre-test escalation (Table S7). On imputation-free ±7-day windows it excludes the null, with $\rho = +0.77$ (bootstrap 95% CI +0.49 to +0.94), positive in 500 of 500 refits, and a top-to-bottom P/N spread of 3.16 (95% CI 1.18–4.95). On the full, interpolated sample the same interval includes zero (+0.49, 95% CI −0.14 to +0.71), so the ordering is established on imputation-free windows and not on the data as a whole.

The stable-dose branch is by contrast almost perfectly reproducible (adjusted Rand index 0.99; per-cluster Jaccard at least 0.98), which we do not read as evidence in its favour, because a density-valley rule applied to a stable density is close to deterministic and will reproduce whether or not the valleys correspond to anything real.

Cluster associations with morphine-positive urine

Applying the positive-derived taxonomy to all tests, cluster membership was associated with the urine result after patient-level correction and false-discovery-rate control (Table 2, Figure 2). Because the taxonomy is defined on positives and could therefore be circular, the associations were re-estimated with outcome-blind clustering of the pooled sample. The two partitions agree at an adjusted Rand index of 0.60, which is not high in absolute terms and is not comparable with the 0.62 bootstrap figure above, so the outcome-blind check is judged on the cluster-by-cluster crosswalk and not on the index.

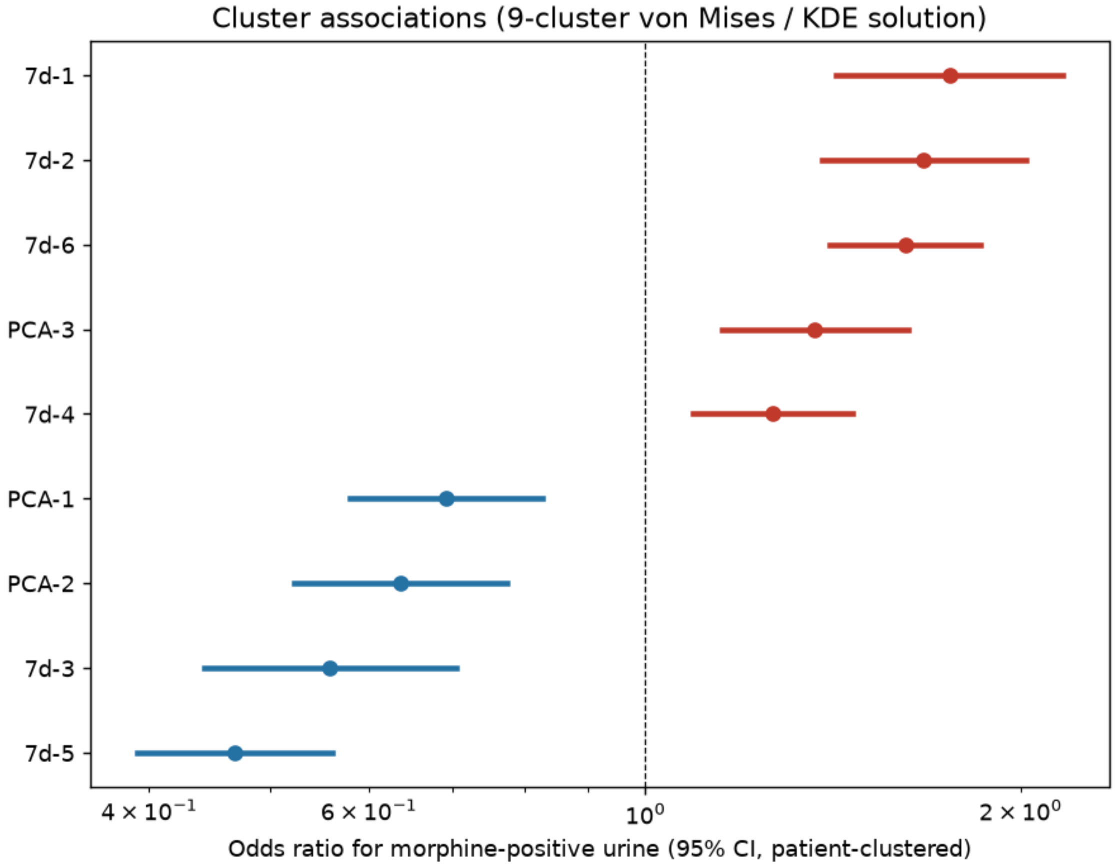

Figure 2. Forest plot of odds ratios with 95% confidence intervals, patient-clustered, by cluster under the positive-derived taxonomy.

The three stable-dose clusters are nonetheless not equally weak, and we distinguish insufficient evidence from contrary evidence. PCA-2 is the strongest, as it alone replicates outcome-blind (P/N 0.65 to 0.69) and holds on the imputation-free subset, although on fully observed ±28-day windows it keeps its protective direction (0.78) without reaching significance. PCA-1 retains its direction throughout (0.77 to 0.84) but fails outcome-blind replication. PCA-3 alone attracts contrary evidence, in that an alternative von Mises partition reverses its direction. We therefore treat PCA-2 and PCA-1 as unconfirmed hypotheses and PCA-3 as contradicted (Tables S1, S2).

Table 2. Clusters ranked by positive/negative occurrence ratio under the positive-derived taxonomy (patient-clustered logistic regression, false-discovery-rate adjusted).

Cluster	Branch	Trajectory (mean pre → post slope)	Positive n	Negative n	P/N	OR (95% CI)	q-value	Label
7d-1	dose-change	rise before, rise after (+0.10 → +0.10)	529	618	1.64	1.75 (1.42–2.16)	3.2×10^{-7}	Higher-risk
7d-2	dose-change	rise before, rise after (+0.05 → +0.10)	262	309	1.62	1.67 (1.38–2.02)	2.7×10^{-7}	Higher-risk
7d-6	dose-change	flat before, rise after (-0.00 → +0.11)	532	669	1.52	1.62 (1.41–1.86)	6.1×10^{-11}	Higher-risk
PCA-3 †	stable-dose	stable dose (-0.00 → -0.00)	267	381	1.34	1.37 (1.15–1.62)	3.9×10^{-4}	Higher-risk (exploratory, not claimed)
7d-4	dose-change	rise before, flat after (+0.11 → -0.00)	428	664	1.23	1.27 (1.09–1.47)	1.7×10^{-3}	Higher-risk
PCA-1	stable-dose	stable dose (-0.00 → -0.00)	908	2240	0.77	0.69 (0.58–0.83)	6.8×10^{-5}	Lower-risk
PCA-2	stable-dose	stable dose (-0.00 → -0.00)	161	471	0.65	0.64 (0.52–0.77)	9.2×10^{-6}	Lower-risk
7d-3	dose-change	flat / slight fall (-0.00 → -0.00)	107	357	0.57	0.56 (0.44–0.71)	1.7×10^{-6}	Lower-risk
7d-5	dose-change	flat / slight fall (-0.00 → -0.00)	207	790	0.50	0.47 (0.39–0.56)	1.2×10^{-15}	Lower-risk

† The three stable-dose clusters (PCA-1, PCA-2, PCA-3) are exploratory for three independent reasons. An alternative von Mises partition of the same tests yields no higher-risk stable-dose cluster; PCA-1 and PCA-3 do not replicate under outcome-blind clustering; and none of the three remains significant when restricted to the 3,737 tests with a fully observed ±28-day window (PCA-1 P/N 0.84, PCA-2 0.78, PCA-3 1.39, all q between 0.07 and 0.14; Supplementary S1). PCA-3 in particular is not claimed, and the observed-collection-negative analysis does not support it. Table S1 gives every cluster under every sensitivity analysis.

Trajectory interpretation

The clusters had clinically coherent shapes (Figures 3 and 4). Because the primary signal is the consumed dose, the higher-risk clusters (7d-1, 7d-2, 7d-6, 7d-4), in which the patient receives and takes away more methadone in an escalation or peri-test rebound, plausibly represent patient behaviour such as dose-seeking or self-medication of withdrawal, together with any clinical dose response, while the lower-risk clusters (7d-5, 7d-3) showed tapering or stability. Two caveats attach to this reading, the first being that the dose-change window straddles the test day, so the contrast describes behaviour around a positive result and should not be read causally. And observed ingestion is not proof of absorption, so the behavioural reading remains an interpretation of a consumption record.

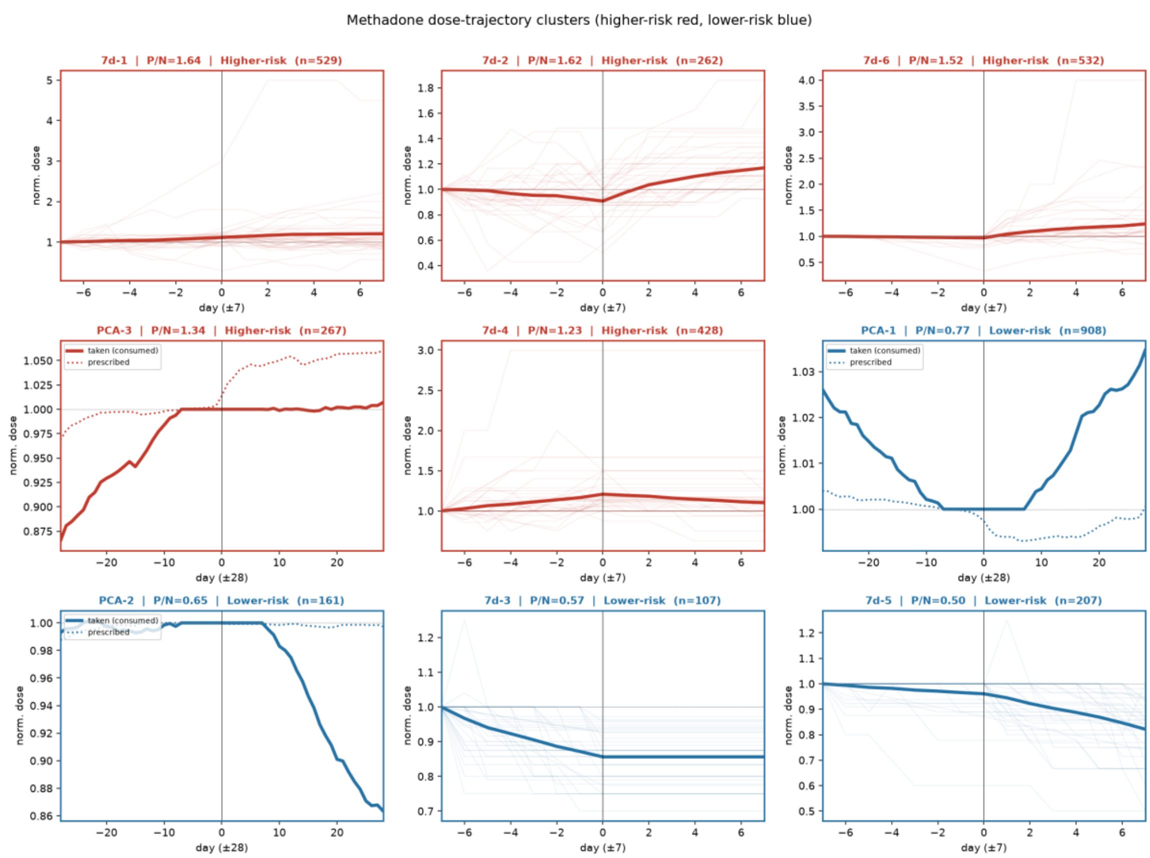

Figure 3. Cluster gallery showing the mean consumed-dose trajectory of each of the nine clusters (higher-risk in red, lower-risk in blue), with thin lines for individual tests and a thick line for the cluster mean. Dose-change clusters are shown over ±7 days and stable-dose (PCA) clusters over ±28 days, with the consumed series solid and the prescribed series dotted.

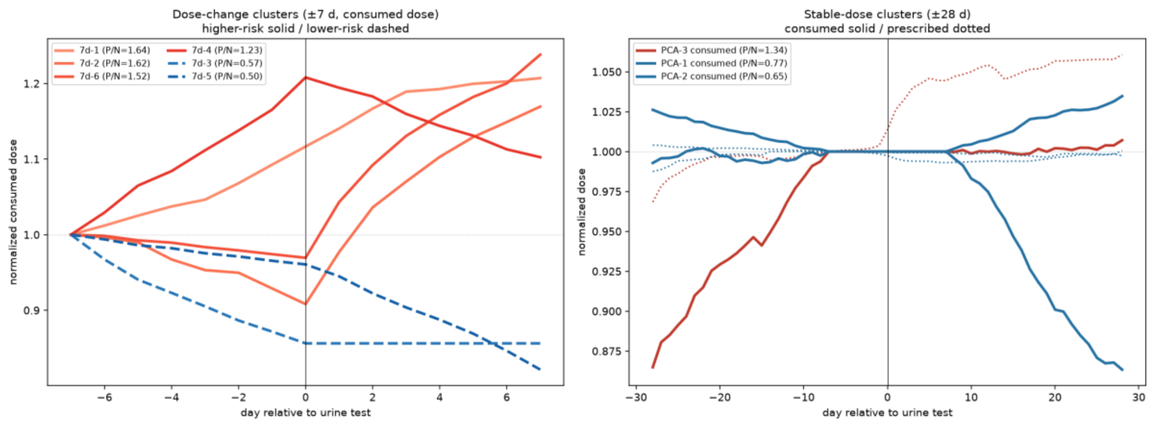

Figure 4. All significant clusters overlaid, with the left panel showing dose-change clusters over ±7 days for the consumed dose (higher-risk solid, lower-risk dashed) and the right panel showing stable-dose clusters over ±28 days (consumed solid, prescribed dotted).

Attendance

Attendance in the week before testing was markedly lower before positive than before negative tests (means 0.81 and 0.91 respectively; SD of the attendance variable 0.24). In a patient-clustered model, higher attendance was strongly protective, with an odds ratio of 0.84 per 0.1 increment ($p = 7\times10^{-22}$; Figure 5). For comparison with the slope features this is 0.65 per standard deviation; expressed over the full range of the variable it is 0.17, and it is that whole-range figure, rather than any clinically meaningful unit, that produces the dramatic-looking number. All attendance estimates below are given per 0.1 increment.

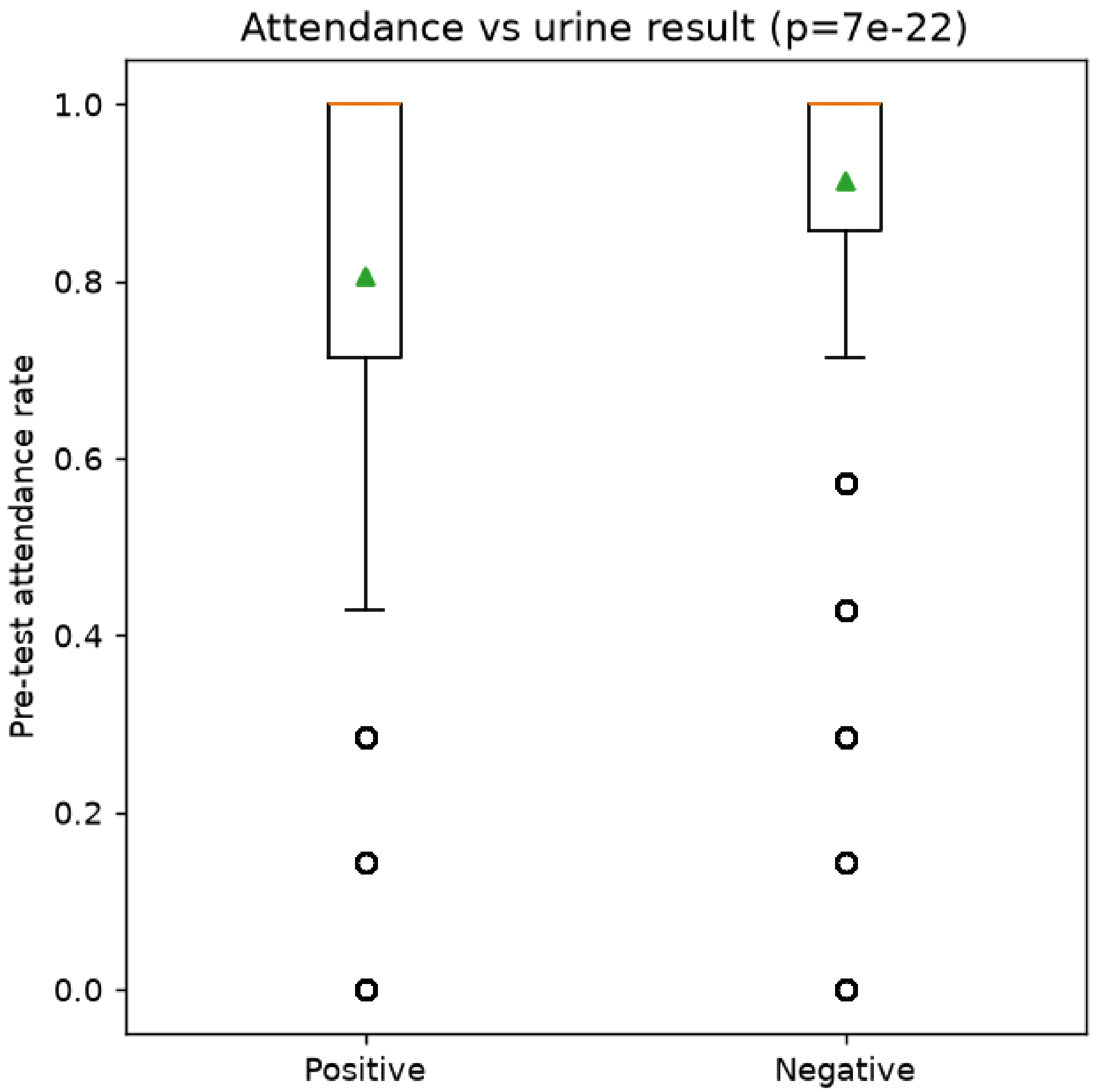

Figure 5. Distribution of pre-test attendance by morphine urine result, over the week preceding each test.

Sensitivity analyses

The analyses that determine what is claimed are summarised here and reported in full in Supplementary S1 and S5, together with the pre-test and post-test decomposition, resolution robustness, the observed-collection negative subset, the duplication check and an independent raw-data reconstruction.

All nine clusters remain significant with pre-test attendance in the model, although two attenuate materially (7d-6, OR 1.62 to 1.29; 7d-2, 1.67 to 1.37). Cluster membership adds decisively to a continuous slope, and does so against the claimed exposure as well as the disclaimed one (likelihood-ratio χ^2 of 328, 261 and 254 on 8 degrees of freedom against the pre-test slope, the post-test slope and both together respectively, all $p < 10^{-49}$). Splitting at the median date of April 2020, the attendance association replicated in the held-out later period (OR 0.86 to 0.78 per 0.1 increment, both $p < 10^{-7}$), whereas the dose-change association did not (earlier OR 1.18, $p = 0.01$; later 1.15, interval crossing unity, $p = 0.17$). The dose finding therefore has no out-of-time replication, a qualification of the primary claim and not merely a limitation.

Re-clustering the pooled sample without reference to the urine result yielded ten clusters. Matching cluster by cluster (Table S2), 6 of the 9 primary clusters are reproduced by a distinct outcome-blind cluster of the same direction and near-identical P/N; 7d-2 has no distinct counterpart, and among stable-dose clusters only PCA-2 replicates. The dose-change associations are thus unlikely to be an artefact of clustering on positives, although they replicate at a coarser resolution than the reported partition. Refitting with patient fixed effects (451 patients contributing both outcomes) leaves the associations undiminished on a common scale, and 8 of 9 clusters were significant within-patient in the direction of Table 2, so these differences appear to be peri-test phenomena and not between-patient case-mix.

Non-attended days were interpolated upstream, so the decisive test restricts attention to windows in which no day was interpolated, defined from the pre-fill series and not from attendance, comprising 5,644 tests over ±7 days (57%) and 3,737 over ±28 days (38%). An earlier definition of this subset was looser and has been corrected (Supplementary S1). Comparing each quantity against its complement gives a double dissociation. The post-test escalation magnitude does not survive. Within patients it falls from 1.36 to 1.06 per standard deviation (95% CI 0.97–1.15, $p = 0.23$) and rises to 1.44 on the complement, so it appears to be largely generated by the imputed days. Pre-test escalation shows the mirror image, strengthening from 1.15 to 1.26 (95% CI 1.15–1.38, $p = 3\times10^{-7}$) against an inconclusive 1.07; it holds within patients, the pooled between-patient estimate being 1.23 (95% CI 0.98–1.56, $p = 0.078$), so the pre-test finding is a within-patient claim. Cluster membership survives, with 7 of 8 contrasts against 7d-1 still significant. The dissociation follows from the representation, in that interpolation inserts a straight segment, inflating the length of the slope vector, whereas the polar-angle features retain only its direction. Because non-attendance both generates the imputed values and is the strongest correlate of a positive result, adjusting for attendance cannot remove this contamination, which lies inside the exposure variable. We therefore claim cluster membership and the pre-test component, and do not claim the post-test escalation magnitude.

A pre-test rise might reflect the clinician prescribing more, since what a patient consumes is bounded by what is prescribed. The prescribed-dose comparator, computed with the same algorithm as the exposure, is only weakly related to it ($r = 0.14$), and adjusting for it leaves the consumed-dose association essentially unchanged. The prescribed dose is nonetheless not inert. On imputation-free windows its own pre-test slope is associated with the result (OR 1.13, 95% CI 1.04–1.22, $p = 0.003$), independently of consumption, so the peri-test window contains a clinical response as well as a patient behaviour that this design cannot separate (Figures 6 and 7; Supplementary S5).

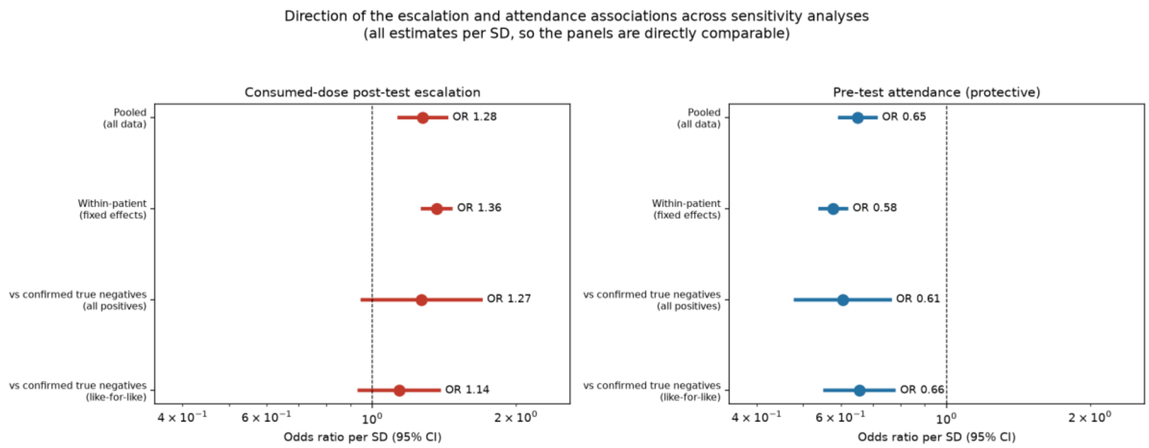

Figure 6. Robustness of the key associations, showing consumed-dose escalation (higher-risk, left panel) and attendance (protective, right panel) odds ratios per standard deviation with 95% confidence intervals across the pooled, within-patient (fixed-effects) and confirmed-true-negative analyses.

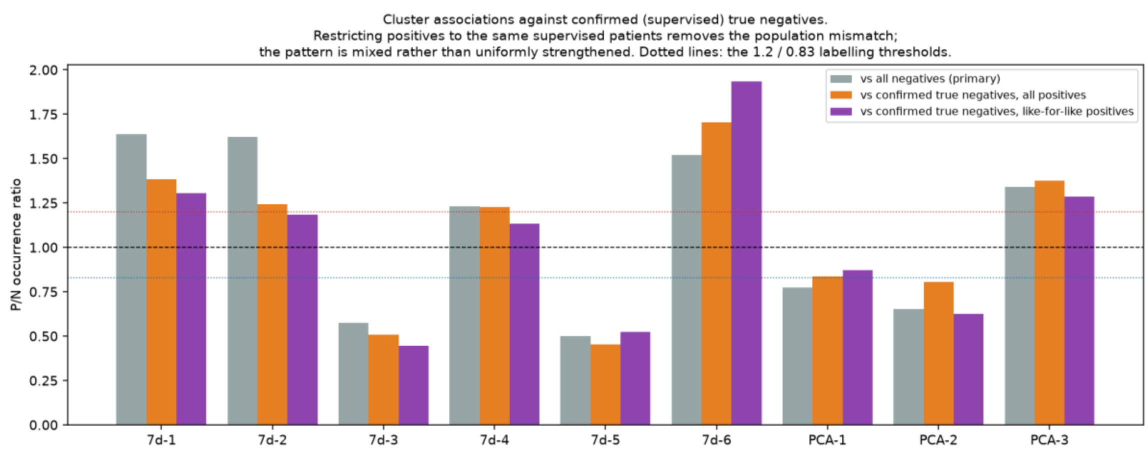

Figure 7. Cluster occurrence ratios (P/N) against all negatives compared with confirmed true negatives from supervised collection. Direction is preserved for every cluster, and the strongest higher-risk cluster (7d-6) and PCA-3 strengthen against clean negatives, which would be consistent with false-negative dilution in the full negative set.

Under patient-grouped cross-validation, a baseline of pre-test attendance plus prior urine-test history reached an area under the curve of approximately 0.83, and adding the pre-test dose-trajectory features changed this by approximately 0.00; trajectory features alone reached approximately 0.60 and attendance alone approximately 0.58. The signal therefore carries no incremental predictive value over routinely available history and attendance, and it was not tuned in the paper's favour, an exhaustive search over richer feature sets having failed to raise the ceiling.

DISCUSSION

In a large maintenance cohort, the shape of consumed methadone dose change around a urine test tracked an escalation-ordered gradient in the concurrent morphine result. The gradient survived outcome-blind re-clustering, adjustment for attendance, restriction to windows containing no interpolated day and a within-patient analysis, so it is unlikely to be an artefact of repeated testing, of defining clusters on positives, or of a static between-patient case-mix. It did not, however, replicate out of time across the 2020 split, which is the most serious outstanding qualification on the finding. The contribution is an interpretable description of the dosing that accompanies a morphine-positive result, enabled by a polar-angle representation that makes the shape of dose change comparable independent of its magnitude, the same property that appears to make it robust to the imputation that defeats magnitude-based summaries.

The direction of the effects is behaviourally interpretable. Higher-risk clusters reflect the patient drinking more methadone on site around the test, plausibly dose-seeking or self-medication of withdrawal from illicit use, while lower-risk clusters show tapering or stability, which would be consistent with a settled maintenance phase. Attendance added a strong, independent protective association, reinforcing treatment engagement as a core marker of stability.

Two features of the programme strengthen this reading. The first is that collection occasions were randomly scheduled and not clinician-triggered, so the pre-test dose pattern cannot be an artefact of staff having decided to test a patient about whom they were already concerned; reverse causation of that kind would otherwise be the most serious threat to an antecedent claim. The second is that methadone was consumed on site with no take-home dosing, so the exposure is observed consumption and not an inference from dispensing, which is unusual in this literature and is what licenses a behavioural reading at all. Against that, because what a patient consumes is bounded by what is prescribed, the peri-test window holds a clinical response as well as a patient behaviour, and this design cannot separate them.

Three boundaries define what the study supports. The first is that it is associational and not predictive, as the dose-change window straddles the test day and the trajectory added essentially nothing to a baseline of attendance and prior testing history. Second, only the shape of dose change, and its pre-test component, are robust to the imputation of non-attended days, which follows from the polar-angle representation discarding the length of the slope vector and requires the post-test escalation odds ratio to be set aside. Third, the stable-dose branch is exploratory and nothing in it is claimed. The study therefore offers a descriptive, hypothesis-generating account of the dosing behaviour that accompanies illicit-opioid use and not a risk-prediction model.

Clinical interpretation

The effect sizes translate into modest to moderate shifts in absolute risk. Taking the analysis-sample positive rate of roughly one third as an illustrative baseline, given that the true per-visit rate is programme-specific, the strongest higher-risk pattern (odds ratio 1.75, Table 2) corresponds to a rise from roughly 34% to 47%, and the protective tapering pattern to a fall from roughly 34% to 20%. The dominant signal is attendance, and it is worth stating on a scale a clinician can use. Each 0.1 increase in the proportion of attended days in the week before the test is 0.7 of a day, or about one extra attended day per ten days, and carried an odds ratio of 0.84; a patient attending 6 of 7 days rather than 4 of 7, an increase of 0.29, therefore has roughly 0.60 times the odds of a positive test. Declining attendance is thus the clearest and most actionable warning sign, and a rising consumed dose relative to the patient's own baseline a useful corroborating signal though not a standalone alarm. These are population-level associations, and not thresholds for flagging an individual, as no single marker predicts the next result with useful accuracy.

Limitations

Urine testing followed routine clinical procedures and under-detects use, and we treated results at face value and clustered on positives to limit contamination of the partition. The supervised-collection subset does not fully solve this. Observation assures specimen validity but leaves the detection window of the assay unchanged, and the cited evidence locates the false-negative problem in testing frequency and not in collection procedure [5,6], which for a cohort attending daily is precisely the scenario in which routine schedules detect only frequent use.

The pipeline was run only on patients retained for at least two years, excluding 5,214 of 15,592 dosing-linked records. The excluded occasions had a higher positive rate and much lower attendance, so these results describe stably retained patients and should not be extended to early treatment, the phase in which a marker of concurrent use would probably be most useful clinically. Non-attended days were interpolated before feature construction, so the dose-change features are partly a function of when a patient did not attend; the subset definitions that bound this, including a correction to an earlier definition, are given in Supplementary S1.

The number of dose-change clusters was chosen by us and not selected by a criterion, and the nine-cluster partition is one of several defensible cuts of a continuum; boundaries move under resampling, so the labels are a descriptive device. The stable-dose branch admits more than one partition with materially different conclusions, and there is no registered protocol for this study. The study is retrospective and single-programme, and the dose-change association did not replicate out of time across the April 2020 split, which the pandemic onset may partly explain but does not excuse; external validation seems necessary before the finding is treated as established. Finally, age and sex were absent from the data extract and cannot be reported (Table 1), so residual confounding by demographic case-mix cannot be assessed.

Conclusions

Patterns of consumed methadone dose around a urine test, captured by a polar-angle partition of dose-change shape, are associated with concurrent morphine-positive results, with escalation or rebound marking higher and tapering or stability marking lower likelihood, alongside a strong protective attendance association. The dose-change patterns replicate outcome-blind and hold within patients, although the post-test component of their magnitude appears to be generated by the imputation of non-attended days and the association did not replicate out of time; the stable-dose patterns are exploratory only. These are best read as behavioural signatures of concurrent illicit-opioid use and not as a predictive tool, and they motivate prospective, multi-centre study.

SUPPLEMENTARY MATERIAL

Supplementary S1 reports the sensitivity analyses in full, covering adjustment for attendance, the taxonomy-versus-single-scalar comparison, the pre-test and post-test decomposition, resolution robustness across K = 4 to 8, temporal validation across the 2020 split, the observed-collection negative subset including the like-for-like restriction, and an independent raw-data reconstruction. Supplementary S5 reports in full the estimates summarised under Sensitivity analyses. The accompanying tables are Table S1 (every cluster under every sensitivity analysis),

Table S2 (the outcome-blind crosswalk and adjusted Rand index), Table S3 (clustering reproducibility from 500 bootstrap refits, which also corrects an earlier version in which the stable-dose figure was taken from a bootstrap of a von Mises mixture and not of the density-valley cut procedure actually used for Table 2), Table S3b (the same intervals after matching each refit's components to the full-data clusters by Jaccard overlap, from a separate 200-refit run that must not be conflated with Table S3), Table S4 (the cluster-outcome pattern at K = 4 to 8, with the K = 4 value flagged as degenerate), Table S5 (the full breakdown of the imputation of non-attended days), Table S6 (the alternative one-dimensional von Mises partition of the stable-dose branch) and Table S7 (the bootstrap distribution of the escalation-P/N rank correlation and of the top-versus-bottom P/N spread, on which the ordering claim rests).

DATA AND CODE AVAILABILITY

Analysis code is available, covering the participant flow (`35_strobe_flow.py`, which re-derives every count in the Results from the source spreadsheets and asserts them against the analysis file), the primary cluster table (`36_final_combined.py`), the clustering-reproducibility metrics (`40_stability.py`), the sensitivity and outcome-blind-crosswalk tables (`37_supp_sensitivity.py`), the reporting-scale, imputation and PCA descriptives (`38_units_descriptives.py`), the resolution check (`39_k_robustness.py`), the corrected imputation-free subsets (`42_correct_imputation_free.py`), the participant-flow linkage and duplication diagnostics (`43_linkage_dedup_checks.py`), the bootstrap gradient statistic (`44_gradient_bootstrap.py`), the complement and duplication sensitivity analyses (`45_complement_and_dedup.py`), the matched bootstrap intervals (`46_audit_bootstrap_and_ci.py`), the alternative stable-dose partition (`48_stabledose_alt_partition.py`), the prescribed-dose adjustment (`49_confound_same_exposure.py`), the likelihood-ratio comparisons

(`50_lrt_and_cluster_shape.py`), the predictive-value analyses (`33_max_predict.py`, `20_pretest_only.py`), the figures (`15_final_figs.py`, `16_cluster_gallery.py`, `25_overlay.py`, `27_more_figs.py`) and the reference checks (`scripts/check_refs.py`, `scripts/audit_ref_format.py`). The upstream construction of the 57-day dose windows and the interpolation of non-attended days were written by the authors in MATLAB (`retrieve_raw_data.m`, `Index_ver2.m`) and are read by, and not re-implemented in, the Python pipeline, and that MATLAB source is included. Data are available subject to institutional and ethical approval.

DECLARATIONS

Ethics approval

The study was approved by [ETHICS COMMITTEE NAME] under approval number [IRB NUMBER], and individual informed consent was waived for this retrospective analysis of routinely collected, de-identified records.

Author contributions

[COMPLETE PER AUTHOR, adding co-authors and their roles. A suggested mapping for the corresponding author, to be edited, is Conceptualization; Methodology; Formal analysis; Data curation; Writing – original draft; Writing – review & editing; Supervision. Every listed author must meet all four ICMJE criteria and must be able to take public responsibility for the content.]

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Declaration of competing interests

[STATE ANY COMPETING INTEREST, or state that the authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.]

Declaration of generative artificial intelligence and AI-assisted technologies

During the preparation of this work the authors used a large language model (Anthropic Claude) to assist with implementation of the analysis code, with the design and execution of the sensitivity analyses reported here and in the Supplementary material, and with drafting and editing the English text. The study question, the polar-angle representation of dose change on which the method rests, the source data and every decision about what is and is not claimed are the authors' own. All statistical output was regenerated from the source records by scripts that the authors retain and can re-run, and the AI-assisted text and analyses were reviewed, verified against those outputs, and corrected by the authors, who take full responsibility for the content of this publication. No AI system is listed as an author, and none met authorship criteria.

Data protection statement

De-identified records (pseudonymous patient identifiers, dates, qualitative urine results and daily dose values, with no names) were processed during AI-assisted analysis, and no directly identifying field was transmitted. [CONFIRM WITH THE ETHICS COMMITTEE THAT THIS PROCESSING IS COVERED BY THE APPROVED PROTOCOL, AND AMEND THIS STATEMENT ACCORDINGLY BEFORE SUBMISSION.]

Reporting guideline

This observational study is reported in accordance with the STROBE statement, and the participant flow is given in the Results.

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